

Yash Sandeep Kokane

Pre-final Year Undergraduate

Dual Major - Materials Science and Computer Science Engineering

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Education

Degree	Institution	CPI/%	Year
B.Tech	IIT Gandhinagar	7.93	2020 - Present
Class XII	PACE Junior College, Thane	85.00	2019-2020
Class X	AK Joshi School, Thane	94.40	2017-2018

Internships

- **Optimization and MPC Modelling for Acoustic Wave Separators | TCS Research** *May - July 2023*
Mentor: Dr. Venkata Buddhiraju
 - Worked on a mechanistic model for Acoustic Wave Separators (AWS), a cutting-edge biopharmaceutical device used for cell culture clarification.
 - Optimized control parameters of the AWS device using Genetic Algorithms to maximize product throughput.
 - Also formulated a Model Predictive Control (MPC) Framework to dynamically determine optimal control parameters, accounting for potential scaled-up industrial implementations.
- **Invention Factory Research Programme by the Cooper Union, New York** *May - July 2022*
 - Collaborated with 40+ professors and entrepreneurs from IITs to conceive novel mechanical and robotic solutions that can positively impact society.
 - Successfully patented "LiftSafe" (Provisional Patent Application Number: 202221042912), a novel portable weightlifting safety device designed to prevent weights from endangering users.
- **Acoustic Model for Marathi language | PinnacleWorks Infotech** *Aug 2023 - Nov 2023*
 - Worked on developing Acoustic Models for Marathi language speech-to-text conversion using the Kaldi Deep Learning toolkit.
 - Scraped over 10 Million sentences of Marathi Language data from Marathi Wikipedia and news channels to develop Marathi Language Models.
 - Mapped phonemes, tuned DNN (Deep Neural Network) hyperparameters to develop a working Marathi speech-to-text model with a Word Error Rate (WER) of 9%.

Projects

- **ML Driven Optimization of Lennard-Jones Parameters for MD Simulations** *Jan 2024*
Mentor: Prof. Anirban Mondal
 - Developed and implemented an approach combining Gaussian Process Regression and Genetic Algorithms for optimizing Lennard-Jones parameters involved in Molecular Dynamics (MD) Simulations, significantly improving accuracy for simulating transport phenomena in Lithium-Ion batteries.
 - Currently working on validating results through extensive Molecular Dynamics simulations using GROMACS.
- **Verification of the Stokes-Einstein Equation and Central Limit Theorem through synthesis and observation of colloidal polystyrene particles. [\[Report\]](#)** *Sept - Nov 2022*
Mentor: Prof. Chandan Mishra
 - Synthesized uniformly sized polystyrene spherical colloidal particles, by preparation of polystyrene seed latex, followed by the seeded growth technique and dialysis to remove impurities.
 - Developed custom algorithms in OpenCV to observe and track motion of synthesized particles under a digital microscope to obtain mean square velocity of particles. Calculated the Diffusion Constant of the colloidal particles using the Stokes-Einstein Equation and Langevin Equation.
 - Verified the Central Limit Theorem with respect to the mean random walk displacement of synthesized particles over time.
- **Team Lead, Analysis of infill parameters involved in 3D Printing. [\[Report\]](#)** *April 2022*
Mentor: Prof. Ravi Sastri Ayyagari
 - As part of the ES 301 (Mechanics of Solids) course, ideated and conducted an in-depth study of an FDM 3D printer to determine optimal infill and material parameters.
 - Conducted tests using CTM Machines, Charpy Impact machines, Rockwell Machines etc. in a team of 5.
 - Utilized Ansys and AutoCAD Fusion for detailed simulations.
 - Won the *first prize among 35+ competing teams* as part of the course project.
- **Identifying crack-inducing parameters in continuous casting process of steel** *Jan 2023*

Mentor: Prof. Amit Arora

- An industrial project offered by ArcelorMittal, worked on identifying crack-inducing parameters associated with the Continuous Casting process of Steel at the Hazira plant of ArcelorMittal-Nippon Steel.
- Conducted exploratory data analysis of time series data from various sensors, feature importances of Random Forest Classifiers were used to pinpoint influential factors.
- Developed a RNN to predict crack propagation in steel slabs.

• Sports Analytics using Computer Vision and Deep Learning*July 2023 - Present***Mentor: Prof. Ravi Hegde**

- Trained custom Yolo detection models for javelins and used them with OpenCV for calculating biomechanical parameters associated with Javelin Throw.
- Used Alphapose and MotionBERT models to generate large 3D body pose datasets of Javelin Throws from 2D videos of throws obtained from Javelin Throw tournaments.
- Working on incorporating Graph Neural Networks for studying throw parameters from 3D poses, to study the effect of these parameters on throw distance.

• Synthesis and characterization of rolled and annealed Ni-Co alloys*Jan - April 2023***Mentor: Prof. Pradipta Ghosh**

- Synthesized and characterized Ni-10Co alloys rolled and annealed at different temperatures to understand the phenomena of recrystallization and annealing.
- Conducted rolling and controlled annealing of Ni-10Co alloy samples, followed by XRD, XRF, EDS and Hardness analysis and observation under SEM to identify twinning and calculate average grain size.

Positions of Responsibility**• Teaching Assistant, MOOC Course on ML in Materials Science***December 2023*

- Selected as a Teaching Assistant for the GIAN MOOC Course conducted by IIT Kharagpur, conducted by Prof. Subramanian Sankaranarayanan (University of Illinois Chicago) and Prof. Raghavan Ranganathan (IIT Gandhinagar).
- Helped participants with doubt resolution after instruction hours, in addition to ensuring smooth logistics of the course.

• Academic Discussion Hours Mentor, IIT Gandhinagar*April 2023 - July 2023*

- Was chosen as an academic guide for conducting out-of-classroom lectures for academically struggling undergraduate students, on the basis of excellent academic performance.
- Academic mentor for 2 courses: Introduction to Materials Science, Materials of the Future.

• Class Representative, Materials Science Department*August 2022 - August 2023*

- Acted as a bridge between classmates and faculty to address academic and administrative matters. Effectively communicated class feedback, fostering collaborative improvements.

• Core member, Mean Mechanics (Robotics Club), IIT Gandhinagar*May 2021 - May 2022*

- Conducted multiple workshops on Robotics Hardware and Arduino Interfacing for the student community of IIT Gandhinagar.

Skill Summary

- **Languages:** Python, C, C++, Javascript, Bash Scripting.
- **Tools:** Quantum Espresso, LAMMPS, GitHub, Ansys, Microsoft Excel, MySQL, Linux, MATLAB, Photoshop, Autodesk Fusion, AutoCAD.

Relevant Courses

- Computational Materials (only student to secure **A+ or 11/10 for excellence**), Advanced Materials (A or 10/10), Metallurgy and Extraction (A or 10/10), Introduction to Materials, Physics of Materials, Thermodynamics and Kinetics, Fluid Mechanics, Materials Processing, Corrosion and Degradation of Materials, Introduction to Data Science.
- Completed the **Deep Learning Specialization courses** by Andrew Ng on Coursera.